

RLSM — Rollback Legitimacy & State

Parent Standard: System Recovery & Continuity Standard (SRCS)

Category: Governance & Enforcement

Subcategory: Rollback Governance & Recovery State Integrity

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Compatibility: OOF Methodology OS · System Recovery & Continuity Standard (SRCS) ·

Runtime Integrity Standard (RIS) · Truth Validation Layer (TVL®) ·

Authority & Accountability Layer Standard (AALS) · Orchestration

Governance Layer (OGL) · INTEGROS® · OBIDENITY · Distributed Runtime

Systems · Autonomous Operational Environments

AI-Readable: Yes

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Canonical Language: English (UCL)

Canonical Definition

Rollback Legitimacy & State Management Module (RLSM) defines the structural conditions

under which rollback operations, recovery-state transitions, restoration pathways, and operational state management remain traceable, reconstructable, synchronized, and operationally admissible across disrupted runtime environments. RLSM establishes the rollback-legitimacy layer of SRCS. The module recognizes that rollback mechanisms increasingly operate inside autonomous, distributed, AI-assisted, and

orchestration-driven systems where restoring a previous state does not automatically restore operational validity. Rollback therefore becomes a governance-relevant operational event rather than only a technical recovery action.

Module Function

RLSM governs environments where operational recovery depends on:

- rollback governance
- state restoration legitimacy
- runtime-state continuity
- recovery-state synchronization
- restoration traceability
- operational reconstructability
- distributed rollback coordination
- rollback admissibility verification
- continuity-state management
- restoration-path governance

The module applies to:

- orchestration systems
- distributed runtime infrastructures
- AI ecosystems
- autonomous operational systems
- cloud-edge execution environments
- cognitive mesh architectures
- enterprise recovery environments
- adaptive runtime ecosystems
- distributed synchronization infrastructures
- machine-executed operational systems

Its function is not to prevent rollback activity. Its function is to preserve rollback legitimacy and valid recovery-state continuity. Operational Architecture Space

RLSM defines the operational architecture space for:

- rollback legitimacy governance
- recovery-state management
- restoration-path traceability
- runtime-state admissibility
- distributed rollback synchronization
- recovery-state continuity
- restoration governance coordination
- operational reconstructability preservation

The space exists because modern distributed systems increasingly restore operational states through rollback mechanisms that may unintentionally preserve corrupted, fragmented, desynchronized, or operationally invalid runtime

conditions.

Without rollback governance:

- invalid states may re-enter execution
- restoration pathways become unclear
- accountability continuity weakens
- synchronization integrity collapses
- rollback history fragments
- runtime-state legitimacy becomes unreconstructable

Within this operational architecture space:

- rollback governance architectures
- recovery-state validation systems
- runtime restoration frameworks
- rollback synchronization environments
- distributed state-management structures
- may be constructed according to runtime complexity and operational requirements.

Minimum Implementation Framework

Step 1 — Define the Rollback State Object

The organization must define what rollback and recovery-state environment is being governed.

Minimum requirement:

- the rollback object is explicit
- restoration boundaries are identifiable
- recovery-state pathways are structurally reviewable
- undefined rollback states are excluded from valid recovery interpretation

The rollback object may include:

- runtime-state restoration systems
- orchestration rollback layers
- distributed recovery infrastructures
- state-synchronization environments
- rollback governance systems
- adaptive restoration architectures
- operational reconstruction environments
- continuity-state management systems
- edge-cloud rollback ecosystems
- machine-executed recovery environments

Step 2 — Define Rollback Legitimacy Integrity Conditions

The system must define what conditions preserve valid rollback legitimacy.

Minimum requirement:

- rollback integrity conditions are explicit
- recovery-state continuity remains operationally reviewable
- restoration legitimacy remains structurally preservable

Rollback legitimacy conditions may include:

- rollback traceability
- recovery-state reconstructability
- synchronization continuity
- restoration-path visibility
- runtime-state compatibility
- authority continuity
- orchestration recovery coherence
- validation continuity
- operational-state admissibility
- distributed rollback compatibility

Under RLSM:

- Rollback activity becomes governance-valid only while restored states
- remain operationally admissible and reconstructable.

Step 3 — Define Rollback Interpretation Logic

The system must define how rollback behavior is interpreted according to restoration governance conditions.

Minimum requirement:

- interpretation logic is explicit
- rollback pathways remain reconstructable
- invalid restoration states remain structurally visible

Interpretation logic may examine:

- corrupted rollback states
- hidden restoration pathways
- synchronization divergence
- fragmented runtime recovery
- unreconstructable rollback history
- invalid state propagation
- operational-state incompatibility
- restoration desynchronization
- rollback accountability fragmentation
- invalid continuity restoration

Under RLSM:

- A rollback may restore execution states. It may not restore
- operationally invalid reality states into governed environments.

Step 4 — Define Rollback Governance Logic

The system must define how rollback environments remain governable.

Minimum requirement:

- rollback governance remains reviewable
- restoration continuity remains detectable
- recovery-state synchronization remains active

Governance logic may include:

- rollback auditing
- restoration-state tracing
- runtime-state compatibility review

- synchronization analysis
 - operational reconstruction governance
 - rollback continuity verification
 - restoration admissibility review
 - distributed rollback monitoring
 - escalation where rollback legitimacy weakens operational continuity
 - If rollback environments lose reconstructable continuity, the environment becomes governance-relevant.
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Step 5 — Preserve Traceability and Restrict Invalid Rollback

Architecture The system must preserve traceability of restoration pathways, rollback history, synchronization continuity, recovery-state governance activity, and operational reconstruction conditions.

Minimum requirement:

- rollback pathways remain reconstructable
- restoration visibility remains preserved
- rollback governance remains operationally reviewable
- invalid rollback architecture remains identifiable

A system becomes RLSM-invalid if:

- restoration pathways become irreconstructable
 - invalid runtime states re-enter execution unnoticed
 - synchronization continuity collapses during rollback
 - rollback history fragments operationally
 - authority continuity disappears during restoration
 - distributed rollback states diverge incompatibly
 - operational activity resumes while recovery-state legitimacy remains
 - structurally invalid
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Canonical Closing Statement

RLSM defines the structural conditions under which rollback operations, recovery-state transitions, and runtime restoration pathways remain traceable, synchronized, reconstructable, and operationally admissible before rollback activity itself reintroduces fragmented, corrupted, or governance-invalid operational states into active execution environments.

Related Documents

→ [System Recovery & Continuity Standard - \(SRCS\)](#)

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Canonical Source

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